BY ORDER OF THE COMMANDER
AIR FORCE MATERIEL COMMAND

DEPARTMENT OF THE AIR FORCE
INSTRUCTION 21-101

AIR FORCE MATERIEL COMMAND
Supplement

ADDENDUM B

26 NOVEMBER 2024

Maintenance

LOGISTICS MAINTENANCE PERFORMANCE INDICATOR REPORTING PROCEDURES

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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This publication implements DAFI 21-101 AFMC Supplement, *Aircraft and Equipment Maintenance Management*. It establishes requirements and provides procedures for reporting aircraft maintenance performance indicators (MPI) on all assigned aircraft. This publication applies to all military and civilian AF personnel assigned to the 96th Test Wing and 412th Test Wing as part of the Air Force Test Center (AFTC). This publication does not apply to the Air Force Reserve Command, United States Space Force, or the Air National Guard units or members. DAFI 21-101_AFMCSUP_ADD_B, LOGISTICS MAINTENANCE PERFORMANCE INDICATOR REPORTING PROCEDURES, prescribes Reports Control Symbol (RCS): AFMC-A4M(M) 9302, Monthly Maintenance Performance Indicators' Report. Ensure that all records created as a result of processes prescribed in this publication are maintained IAW Air Force Instruction 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. This publication may be supplemented at any level but route all direct supplements to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("**T-0**, **T-1**, **T-2**, or **T-3**") number following the compliance statement. See DAFMAN 90-161, *Publishing Processes and Procedures*, for a description of the authorities

associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items via regular mail to, Headquarters Air Force Materiel Command Maintenance Division (HQ AFMC/A4M), 4375 Chidlaw Road, Suite C114, Wright Patterson AFB OH 45433, or e-mail to **AFMC.A4M.Workflow@us.af.mil.** Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. For questions on interpreting this instruction, first contact your Quality Assurance Office/activity. Send questions to the OPR at AFMC Maintenance Division (AFMC/A4M) Workflow (**AFMC.A4M.Workflow@us.af.mil**), 4375 Chidlaw Rd, Bldg 262 Room C117 WPAFB OH 45433-5006.

SUMMARY OF CHANGES

This publication has been created to meet the intent of DAFI 21-101, *Aircraft and Equipment Maintenance Management,* and the AFMC Supplement; review this addendum in its entirety. This addendum clarifies mandatory reporting requirements listed in the 9302 report for all wings reporting to AFMC and covers all report templates, detailed requirements for data submission, and training. Requires units to provide explanations for all missed indicators each month, monthly executive summaries, and quarterly health of fleet review to the HQ AFMC/A4M.

Chapter 1

GENERAL INSTRUCTIONS

1.1. Introduction. This instruction, coupled with regular internal performance reviews by both units and headquarters, supports the AFMC Mission — "Powering the world' greatest Air Force…We develop, deliver, support, and sustain war-winning capabilities", by meeting AFMC maintenance performance indicators in Operations, Logistics, and Support programs. This instruction defines logistics performance terms and has reporting and review procedures to enable AFMC to manage by fact. The focus is measurement of the many logistics processes that provide test performance capability to the unit. The result of compliance with this instruction should be the quick and accurate identification of areas for improvement, as well as identification of support problems beyond the scope of the unit. The unit's role emphasizes in-depth analysis of work processes, integrity in measurement methods, timeliness in reporting, and comprehensive remarks describing particular unit support issues requiring further analysis and action. The headquarters' role is to assist AFMC units in process analysis, identification of opportunities for process improvement, identification of process standards and benchmarks, and maintenance of guidelines for consistent unit performance measurement and reporting. This program is called Maintenance Performance Indicators (MPI). By no means should units limit their performance reviews to only the items reported to the headquarters. There are many other metrics available to unit managers that are helpful in determining unit health such as: cannot duplicate (CND) rates, data integrity, deferred discrepancy rates, maintenance scheduling effectiveness, system reliability/capability, attrition, etc.

1.2. Applicability. This instruction applies to all military and civilian AF personnel assigned to the 96th Test Wing (96 TW) and 412th Test Wing (412 TW) as part of the Air Force Test Center. Those AFMC units dependent on Contractor Logistics Support (CLS) and Government/CLS adhere to this instruction to the greatest extent possible within the current contract requirements. AFMC units identify CLS limitations and request waivers to the applicable portions of DAFI 21-101, the AFMC supplements and addendum from HQ AFMC/A4M. AFMC policy is to reduce these waivers, where possible, when CLS contracts are reviewed and/or renegotiated. Write requirements to include logistics support in compliance with this instruction. All contractor requirements contained within this publication must be contained within the contract/grant/agreement to be enforceable.

1.3. Responsibility. Wing and maintenance group commanders are responsible for compliance. Commanders or their designated representative ensure all reports cited in this instruction are prepared and transmitted as prescribed. The preparing agency OPR is the Maintenance Management Analysis (MMA) section (or designated contractor where required) regardless of organizational structure. Commanders, to include Operations Group and sub-units, review the accuracy of data required by this instruction and take action to improve deficiencies. At a minimum, the Maintenance Operations Superintendent has overall responsibility for ensuring quality, in-depth analysis of drivers in the Executive Summary.

1.4. Reporting During Minimize Conditions. Minimize conditions are controlled by a designated communication agency. A minimize condition occurs to limit message traffic during contingencies as needed. All reporting continues during minimize conditions with "Minimize

Considered" entered at the bottom of each page of the report. Reports cited under this instruction are essential and require expeditious preparation and transmission during emergency conditions.

1.5. Algorithms. Formulas used in computing rates and performance indicators are listed in **Attachment 2** of this publication

1.6. References. Personnel responsible for the reports described in this instruction should be familiar with the applicable portions of T.O. 00-20-2, *Maintenance Data Documentation*, AFI 10-601 *Operation Capability Requirements Documentation and Validation*, DAFI 21-103, *Equipment inventory, Status and Utilization Reporting,* DAFI 63-101/20-101, *Integrated Life Cycle Management*, AFCSM 21-556 V2, *Intro to IMDS CDB,* AFCSM 21-558, *Comprehensive Engine Management System (CEMS)*, AFCSM 21-564, *Status and Inventory Reporting,* AFCSM 21-574, *Automated Debriefing,* AFCSM 21-579, *Maintenance Supply Interface,* DAFI 21-101, and DAFI 21-101_AFMCSUP.

Chapter 2

9302 REPORT TRANSMISSION INSTRUCTIONS

2.1. Report Transmission Methods. Submit the Monthly Maintenance Performance Indicators Report, RCS: AFMC-A4M 9302 via the HQ AFMC/A4MM SharePoint Site at **https://usaf.dps.mil/teams/40368/TW_Metrics/Forms/AllItems.aspx**. Contact HQ AFMC/A4MM for assistance at DSN 574-3802 or email the organizational box.

2.2. Report Templates. Each wing/unit will work with HQ AFMC/A4MM on their respective 9302 report template specifically designed for its Mission Design Series (MDS) and reporting requirements. **(T-2)** For example, the template used for the 96 TW is named "0096TW-9302-TEMPLATE.XLSM". Use of the templates is mandatory. **(T-2)** When the file is saved locally, DO NOT change the file extension of the template file, keep as .XLSM always throughout the process. Each month, replace the word "TEMPLATE (X.X)" in the file name with the current month (3 characters) and year (4 digits). Example: For March 2016, the 96 TW file name would be "0096TW-9302-MAR-2016.XLSM". Wings/Units should check this site every month before reporting to ensure they have the current 9302 for their unit. The version of the file is indicated at the end of the file name (e.g. 0096TW-9302-MAR-2016(3.4).XLSM).

2.2.1. Template Modifications. Do not modify the template without coordination from HQ AFMC/A4M. If changes are required, contact the HQ AFMC/A4MM, Air Force Materiel Command Maintenance Management Branch to initiate a change (DSN 787-3802).

2.3. Restriction on Types of Information. Only UNCLASSIFIED information is acceptable. Do not include CLASSIFIED information in the report. If classified information is required on any report, specify an appropriate transmission method (usually SIPRNET e-mail). Call AFMC/A4MM for the current SIPRNET account.

Chapter 3

AIRCRAFT REPORTING REQUIREMENTS

3.1. Types of Reports. This chapter describes the overall report concept. It provides information for each report: Monthly Maintenance Performance Indicators Report, AFMC-A4M(M)9302 and the Monthly Executive Summary to the AFMC/A4/10, Directorate of Logistics, Civil Engineering, Force Protection, and Nuclear Integration.

3.2. Monthly Maintenance Performance Indicators (MPI) Report, AFMC-A4M(M)9302.

3.2.1. Report Use. Intermediate command and Major Command (MAJCOM) managers use this report. It is the official source document for MPI briefings when required for the AFMC Commander and the AFMC/A4/10. It consists of two primary sections: Section 1 (wing overall), Section 2 (drivers).

3.2.2. Report Auto-Calculations and Manual Entries. The data for the 9302 report is manually input. The following rates are automatically calculated: Mission Capable (MC), Total Not Mission Capable Maintenance (TNMCM), Total Not Mission Capable Supply (TNMCS), Test Scheduling Effectiveness (TSE) Deviations, Mission Effectiveness (ME), Sortie Generation, Mx/Ops Deviations, Total Abort, Break, 8/12 Hour Fix, Cannibalization, Repeat, Recur, and Repeat/Recur.

3.2.3. Applicability. Submit this report on all aircraft or systems by MDS. For units possessing more than one MDS, break each out separately. Separate messages are not required for each MDS.

3.2.3.1. Units possessing a mix of F-16 aircraft, F-16C/D series are considered one MDS for 9302 reporting purposes.

3.2.3.2. F-15 MDS includes all variations (i.e., F-15C/D) except F-15E for 9302 reporting. Report F-15E aircraft separately.

3.2.4. Frequency of Reporting. The 9302 report will be submitted to HQ AFMC/A4MM monthly. **(T-2)** Upload the 9302 via the HQ AFMC/A4MM SharePoint site Not Later Than (NLT) 1400 EST on the tenth calendar day following the month being reported. Inform AFMC/A4MM as soon as a situation exists that may cause a late submittal. Exercises and inspections are not valid reasons for late submittals.

3.2.5. Report Preparation. The standard report template is specifically designed for each test wing. Use of the standard report template is mandatory. **(T-2)** Do not deviate from the numbering system outlined in this instruction or in the template.

3.2.5.1. SECTION 1. Section One contains the overall wing/MDS/Block data totals, including contingency information. Section One of each unit's template may contain multiple MDS/Blocks/Missions.

3.2.5.1.1. Specific Template Guidance. The current reporting month-year MUST be entered on this sheet.

3.2.5.2. SECTION 2 – Data Elements/Drivers. Section Two contains the top five drivers (three-digit Work Unit Code (WUC) for TNMCM, TNMCS, Break, Fix, Abort,

Cannibalization, Repeats, Recurs, and ME. Some categories (TSE, ME) can include drivers that are not maintenance driven with no WUC connected.

3.2.5.2.1. Specific Template Guidance. Report the top five drivers for this section monthly.

3.2.5.2.2. For TNMCS drivers and high cannibalization items, provide the full National Stock Number (NSN), part number (if obtained through CLS contractor), full work unit code (WUC) or logistics control number (LCN), and nomenclature. This enables A4MM analysts and AFMC Logistics Readiness Squadron (LRS) personnel to address the correct parts problems. The comment ―NSN not available is not acceptable. Ensure components are listed under the correct subsystem/WUC/LCN.

3.2.5.2.3. For all explanations, use the full WUC/LCN when discussing a part and the three-digit WUC/LCN when discussing the sub-system. For example, the E-3 Anti-Skid Device should be reported as 13DHM, not by the system WUC of 13000. Spell out all acronyms.

3.3. Monthly Executive Summary.

3.3.1. Report Use. Intermediate command, Weapon System Teams, MAJCOM managers, and HQ AFMC/A4/10 use this report. This summary is designed to be a senior leader analysis of major issues/factors that are affecting the unit health. It is designed to be a direct communication between the MXG/CC and the A4/10 Director; however, the information is not privileged.

3.3.2. Applicability. The MXG/CC or equivalent submits the summary on all aircraft assigned to that unit. A designated appointee may sign/submit the summary in absence of the MXG/CC. Separate summaries are not required for each MDS.

3.3.3. Frequency of Reporting. The executive summary is submitted to HQ AFMC/A4MM monthly. Upload the signed summary via the HQ AFMC/A4MM SharePoint NLT 1400 EST on the 12th calendar day following the quarter being reported. Inform AFMC/A4MM as soon as a situation exists that may cause a late submittal. Exercises and inspections are not valid reasons for late submittals.

3.3.4. Report Preparation. The executive summary Senior Leader Comments are not an aircraft-by-aircraft laundry list of issues. Provide a top-level view of topics for the unit. Sample topics are overall fleet health, unit concerns, future plans, Flying Hour Program factors/issues, and MAJCOM required assistance. Concentrate on those issues that headquarters can assist in providing resolutions or eliminating barriers, i.e., supply constraints, reliability concerns, manning or depot related issues (this list is not exhaustive and serve as a few examples). Each MDS in the summary is a separate topic and addressed within the same summary.

3.3.4.1. Provide explanations for major issues each quarter. Comments are not complete unless Who, What, When, Where, Why, and How of the indicator are covered. Do not simply repeat numbers and basic data for the summary. Comments provide important details explaining why an item drove unit performance. Utilize raw data to emphasize a point rather than reiterating information already available in Section 1. Submit thorough comments.

3.3.4.2. Analyze the data. Indicate and note short-term (six months) and long-term (6-24 months) trends. Do not go beyond 36 months. Do not just focus on the current quarter's data to determine if a trend exists. Look at the entire picture to make those determinations.

3.3.4.3. When possible, quantify an issue or event's impact on mission capability, (e.g., one-time inspection reduced aircraft availability by 10%). Although the statement "No Trends Noted" is completely valid, do not use it lightly. Avoid using individual aircraft unless used to quantify specific problems.

3.3.4.4. The unit's report should include comments from maintenance management/sortie production personnel, (i.e., Maintenance Group/Deputy Group Commander, Maintenance Operations, Aircraft Maintenance Unit (AMU) Officer in Charge (OIC)rs, Aircraft Maintenance Squadron (AMXS) Commanders, and Maintenance Operations Officers).

3.3.4.5. List pacing aircraft, limited manning, and unit events such as training, exercises, stand-downs, or deployments that may have affected unit performance. Provide projected unit plans to fix pacing items.

3.3.4.6. Explain the way ahead. One way to accomplish this is to provide "predictive analysis" which means determining where indicators are headed over next several months. Predictive analysis includes, among other things, identifying anticipated changes in aircraft availability, (e.g., expected depot inductions, depot field teams, extended scheduled maintenance, or any other issues that may impact the unit's ability to meet mission requirements).

3.3.5. Report Length. The executive summary should be kept to one page and be concise and reviewed for clarity. Spell out all acronyms.

3.4. Coordination and Correction Procedures. Unit commanders establish coordination procedures to ensure the on-time release of accurate data to HQ AFMC/A4MM. Route the reports through all appropriate maintenance management/sortie production personnel for their review and comments prior to submission to HQ AFMC/A4MM. After the 9302 Data and Executive Summary have been uploaded, submit any corrections to reports via HQ AFMC/A4MM SharePoint with a "heads-up" by e-mail with reference to the original reported data as well as the corrected data. The unit OPR maintains a master file of monthly reports for the current and previous fiscal year. HQ AFMC/A4MM contacts the unit OPR if there are any reporting errors identified after the reports arrive. The OPR, in turn, notifies the appropriate base agency to correct erroneous data. Wing Analysis sections ensure the timely submission of corrected reports.

LINDSAY C. DROZ, Colonel, USAF Director of
Logistics, Civil Engineering and Force Protection

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFCSM 21-556 V2, *Intro to IMDS CDB,* 01 June 2019

AFCSM 21-558, *Comprehensive Engine Management System (CEMS)*, 31 March 2024

AFCSM 21-564 , *Status and Inventory Reporting,* 01 February 2023

AFCSM 21-574, *Automated Debriefing,* 23 September 2022

AFCSM 21-579 , *Maintenance Supply Interface,* 15 September 2022

DAFI 21-101, *Aircraft and Equipment Maintenance Management,* 16 January 2020

DAFI 21-101 AFMC Supplement, *Aircraft and Equipment Maintenance Management,* 18 June 2024

AFI 10-601, *Operation Capability Requirements Development,* 27 April 2021

DAFI 21-103, *Equipment Inventory, Status and Utilization Reporting,* 01 November 2022

DAFI 63-101/20-101, *Integrated Life Cycle Management,* 16 February 2024

AFI33-322, *Records Management and Information Governance Program* 23 March 2020

DAFMAN 90-161*, Publishing Processes and Procedures, 18 October 2023*

T.O. 00-20-2, *Maintenance Data Documentation,* 22 July 2021

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication,* 15 April 2022

Abbreviations and Acronyms

AFKS—AF Knowledge Service

AMU—Aircraft Maintenance Unit

AMXS—Aircraft Maintenance Squadron

CLS—Contractor Logistics Support

CND—Cannot Duplicate

DAFMAN—Department of the Air Force Manual

FCF—Function Check Flight

HHQ—Higher Headquarters

IMDS—Integrated Maintenance Data System

LCN—Logistics Control Number

LRS—Logistics Readiness Squadron

MAJCOM—Major Command

MC—Mission Capable

MDS—Mission Design Series

MMA—Maintenance Management Analysis

MICAP—Mission Capable

MPI—Maintenance Performance Indicators

ME—Mission Effectiveness

MX—Maintenance

MXG—Maintenance Group

NLT—Not Later Than

NMC—Not Mission Capable

NMCB—Not Mission Capable Both

NMCM—Not Mission Capable Maintenance

NMCS—Not Mission Capable Supply

NSN—National Stock Number

OCF—Operational Check Flight

OIC—Officer in Charge

OPR—Office of Primary Responsibility

OPS—Operations

PAA—Primary Aircraft Authorized

PRD—Pilot Reported Discrepancy

RCS—Report Control Symbol

REMIS—Reliability and Maintainability Information System

R/R—Repeat/Recur

TNMCM—Total Not Mission Capable Maintenance

TNMCS—Total Not Mission Capable Supply

TNG—Training

TSE—Test Scheduling Effectiveness

TW—Test Wing

UTE—Utilization Rate

WUC—Work Unit Code

Office Symbols

AFMC/A4/10—Directorate of Logistics, Civil Engineering, Force Protection, and Nuclear Integration

HQ AFMC/A4M—Air Force Materiel Command Maintenance Division

HQ AFMC/A4MM—Air Force Materiel Command Maintenance Management Branch

AFTC/96 TW—Air Force Test Center, 96thTest Wing

AFTC/412 TW—Air Force Test Center, 412thTest Wing

Terms

Deviation—Departure from the printed weekly flying schedule.

Fix Rate—This is a leading indicator showing how well the repair process is being managed. It is a percentage of aircraft sorties with a landing status code of 3 (includes system cap codes 3 and 4) returned to a flyable status in a certain amount of time (clock hours). Refer to AFI 21-103 and MAJCOM directives for maintenance start time, e.g. engine shut down, first chock. Problems found by maintenance after the aircraft lands (ground found) are not considered in the fix time. The fix time stops when all Landing Status Code 3 with system capability code 3/4 Pilot Reported Discrepancies (PRDs) are fixed even if the aircraft remains NMC. This metric is used to track dead time in aircraft repair processes because it measures the speed of repair and equipment maintainability. This includes breaks and fixes that occur at any location, e.g. off-station.

Ground Abort—Event after crew show time that prevents a crew ready aircraft from becoming airborne. Ground aborts are categorized as maintenance (GAA, GAB, GAC), operations, HHQ, weather, sympathy, other, etc. A ground abort for the same discrepancy during an Function Check Flight (FCF)/OCF is not counted in the ground abort rate. If the aircraft ground aborts on a different component during the FCF/OCF, it is counted in the ground abort rate.

Ground Abort Rate—The number of ground aborts (including spare ground aborts) per sortie attempts. This rate includes only ground aborts caused by Maintenance, Operations, and Supply. Ground abort rates can gauge both aircraft reliability and quality of maintenance performed. A ground abort for the same discrepancy during an FCF/OCF is not counted in the ground abort rate. If the aircraft ground aborts on a different component during the FCF/OCF, it is counted in the ground abort rate.

Mission Capable—Expedited backorders in the supply system when material is not available for issue and required by the customer to satisfy mission requirements in a short or expedited period of time. Expedite backorders are appropriate when needed material causes the customer a work stoppage or negatively impacts the mission. A 1A MICAP is for an NMCS condition which is reportable in Section Two of the MPI 9302 report. A JA MICAP is for PMCS conditions.

Not Mission Capable Both Rate—The percent of possessed time an aircraft/system is not mission capable due to both maintenance and supply.

Not Mission Capable Maintenance Rate—The percent of possessed time an aircraft/system is not mission capable due to maintenance.

Not Mission Capable Supply Rate—The percent of possessed time an aircraft/system is not mission capable due to supply.

Operational Check Flight—The first flight of an aircraft that has had extended downtime or extensive maintenance at the request of maintenance to validate a maintenance action.

Possessed Aircraft—Aircraft under a wing commander's operational control and responsibility IAW AFI 21-103, Equipment Inventory, Status and Utilization Reporting.

Recur Rate—A recurring discrepancy on an aircraft occurs on the second through fourth sortie or attempted sortie after corrective action has been taken and the system or subsystem is used. The discrepancy indicates the same malfunction as the original discrepancy and was cleared by maintenance, including CNDs/no-defect-noted, etc. to be classified as a recur. A high recur rate may indicate lack of thorough troubleshooting; inordinate pressure to commit aircraft to the flying schedule for subsequent sorties; or a lack of experienced, qualified, or trained technicians. Examine each recur discrepancy and look for trends.

Repeat Rate—A repeat discrepancy on an aircraft occurs on the next sortie or attempted sortie after corrective action has been taken and the system or subsystem is used. The discrepancy indicates the same malfunction as the original discrepancy and was cleared by maintenance (including CNDs/no-defect-noted, etc.) to be classified a repeat. A high repeat rate may indicate lack of thorough troubleshooting; inordinate pressure to commit aircraft to the flying schedule for subsequent sorties; or a lack of experienced, qualified, or trained technicians. Examine each repeat discrepancy and look for trends.

Repeat/Recur Discrepancy Rate—This metric is a leading indicator. It is the total number of repeat and recur system malfunctions compared to the total number of aircrew discrepancies. A high Repeat/Recur (R/R) rate may indicate lack of thorough troubleshooting; inordinate pressure to commit aircraft to the flying schedule for subsequent sorties; or a lack of experienced, qualified or trained technicians. The more complex the weapon system and the greater the operations tempo, the more susceptible a unit is for repeat or recurring discrepancies. Examine each R/R discrepancy and look for trends. The goal should be to keep all repeat and recurring discrepancies to a minimum.

Scheduled Sortie—An aircraft scheduled for flight by tail number on the weekly flying schedule and confirmed on the daily flying schedule. Incentive flights are considered scheduled sorties and published in the weekly schedule. FCF and OCF are excluded.

Sortie—A sortie begins when an aircraft becomes airborne or takes off vertically from rest at any point of support. It ends when the aircraft returns to the surface (except for continuation sorties in MAF) and meets one of the following: the engines are stopped, or the aircraft is on the surface for more than five minutes, or a change is made in the crew.

Sortie Attempts—Includes sorties flown and ground aborts.

Sortie Generation Rate—The average successful sortie/ground test by specific generation category.

Sortie UTE Rate—The average sorties flown per authorized/chargeable aircraft per month.

Spare Ground Abort—Event after crew show time that prevents a crew ready aircraft from becoming airborne but is replaced by a spare that meets the mission requirement. Spare ground aborts are categorized as maintenance, e.g. GAA, GAB, GAC. The difference between a ground

abort and a spare ground abort is the scheduled line is accomplished, where the ground abort is not scheduled. A spare ground abort is not a deviation but can cause a deviation such as late take-off. Spare ground aborts count in the Ground Abort Rate but do not count towards TSE.

Test Scheduling Effectiveness Rate—This is a leading indicator used to measure how well the unit planned and executed the daily flying schedule (flight and ground tests). The TSE rate is the percentage of sorties/tests flown/ran as scheduled. It also indicates unit turmoil caused by flying schedule deviations. The flying schedule developed by tail number is the baseline upon which the TSE is derived by comparing each day's deviations. Deviations that decrease the TSE from 100 percent include scheduled sorties not flown because of maintenance, supply, operations, weather, higher headquarters (HHQ), air traffic control, sympathy, or other reasons; early takeoffs, late takeoffs, and adds. Disruptions to the flying schedule can cause turmoil on the flightline, sending a ripple effect throughout other agencies, and adversely impact scheduled maintenance actions (refer to AFI 21-101_AFMC Supplement 1 Attachment 8).

Total Abort Rate—This is a leading indicator of aircraft reliability. It is the total number of aborts (air, ground and spare ground) per sortie attempts. Air/ground aborts for the same discrepancy during an FCF/OCF are not counted in the total abort rate. If the aircraft air/ground aborts on a different component during the FCF/OCF, it is counted in the total abort rate. The number of air and ground aborts should match those used for the individual Air and Ground Abort Rates.

Total Not Mission Capable Maintenance Rate—The percent of possessed time an aircraft/system is not mission capable due to maintenance (NMCM) and not mission capable for both maintenance and supply (NMCB). Though a lagging indicator, the TNMCM rate is perhaps the most common and useful metric for determining if maintenance is being performed quickly. It is the average percentage of possessed aircraft (calculated monthly/annually) that are unable to meet primary assigned missions for maintenance reasons (excluding aircraft in B-Type possession identifier code status). Any aircraft that is unable to meet any of its wartime missions is considered Not Mission Capable (NMC). Maintenance managers should look for a relationship between other metrics such as R/R, Break Rate, and Fix Rate to the TNMCM Rate. A strong correlation could indicate heavy workloads (e.g., people are over tasked), poor management, training problems or poor maintenance practices. TNMCM is also called "out for maintenance".

Total Not Mission Capable Supply Rate—The percent of possessed time an aircraft/system is not mission capable due to supply (NMCS) and not mission capable for both maintenance and supply (NMCB). Though this lagging metric may seem a LRS responsibility because it is principally driven by availability of spare parts, it is often directly indicative of maintenance practices. For instance, maintenance can keep the rate lower by consolidating feasible cannibalization actions to as few aircraft as practical. This monthly/annual metric is the average percentage of possessed aircraft that are unable to meet primary missions for supply reasons. TNMCS is based on the number of airframes out for 1A MICAP parts that prevent the airframes from performing their mission (NMCS is not the number of parts that are 1A MICAP). Maintenance managers closely monitor the relationship between the Cannibalization Rate and TNMCS. TNMCS is also called "out for supply".

Utilization Rate—The average number of sorties or hours flown per authorized/chargeable aircraft/system per month. See the Hourly and Sortie Utilization Rate (UTE) Rate Definitions. The UTE rate is a leading indicator but serves as a yardstick for how well the maintenance organization supports the unit's mission. The UTE rate is the average number of sorties or hours flown per

Primary Aerospace vehicle Authorized (PAA) per month as found in the HAF/A3O PB documents. This measurement is primarily used by operations in planning the unit's flying hour program. Maintenance uses this measurement to show usage of assigned aircraft. Since UTE rates are used for planning, actual UTE rates (computed at the end of the month) are used to evaluate the unit's monthly accomplishment against the annual plan. Typically, Combat AF (CAF) units measure the sortie UTE rate, while Mobility AF (MAF) units measure the hourly UTE rate to more accurately measure the combined performance of operations and maintenance.

Attachment 2

ALGORITHMS FOR LOGISTICS - MAINTENANCE PERFORMANCE INDICATORS

A2.1. Maintenance Management Metrics. Metrics provide a measurement of performance and capability. Accurate and reliable information is needed by leaders, supervisors and technicians to make decisions. Primary concerns of maintenance managers are how well the unit is meeting mission requirements, how to improve equipment performance, identifying emerging support problems, and projecting future trends. Maintenance management metrics—sometimes called maintenance performance measures or indicators—are a crucial form of information used by maintenance leaders to improve the performance of maintenance organizations, equipment and people when compared with established goals and standards. Metrics often take the form of an "MC Rate Graph" or a "Status of Personnel Training Slide," presenting a gauge of an organization's effectiveness and efficiency. Properly used, metrics are roadmaps that help determine where the unit has been, where the unit is going, and how (or if) the unit is going to get there.

A2.1.1. The overarching objective of AF maintenance is to maintain aircraft and equipment in a safe, serviceable and ready condition to meet mission needs. Maintenance management metrics serve this overarching objective and are established or maintained by Headquarters AF, Major Commands, Wings and/or Squadrons to evaluate/improve equipment condition, personnel skills and long-term fleet health. Metrics are used at all levels of command to drive improved performance and adhere to well-established guidelines. Metrics should be:

A2.1.1.1. Accurate and useful for decision-making.

A2.1.2. Consistent and clearly linked to goals/standards.

A2.1.3. Clearly understood and communicated.

A2.1.4. Based on a measurable, well-defined process.

A2.2. Analysis is crucial to improving organizational performance and is the key component of the metrics management process. Commanders and maintenance managers should properly evaluate maintenance metrics and rely upon the maintenance analysis section for unbiased information. Analysis sections draw upon information from various maintenance information systems for data. The Integrated Maintenance Data System-Central Database (IMDS-CDB), Field Maintenance Command and Control (FMxC2), Reliability and Maintainability Information System (REMIS), Integrated Logistics System - Supply (ILS-S), AF Knowledge Services (AFKS), Theater Integrated Combat Munitions System (TICMS), and AF/A4-approved command-unique analysis tools are the primary data sources. A good maintenance manager does not strive to chase numbers for the sake of looking good. However, the manager uses metrics to focus resources and personnel to improve maintenance processes. It is critical for managers to clearly understand and communicate the crucial linkage between goals, standards and metrics. The AF sets goals and standards for organizations, personnel and weapons systems that facilitate evaluation, comparisons and improvements. These standards are published separately by senior leadership and should be clearly understood at all levels of command. Leaders at every level support analysis and review metrics to properly drive improved performance. Maintenance analysts manage and track this process, but maintenance metrics, and the resulting improvements they drive, are inherently a leadership responsibility.

A2.3. Primary Maintenance Metrics. Metrics are often grouped into various categories, including leading or lagging indicators. Leading indicators show a problem first, as they directly impact maintenance's capability to provide resources to execute the mission. Lagging indicators follow and show firmly established trends. Maintenance leaders review sortie production and maintenance health constantly and be knowledgeable about maintenance indicators that highlight trends before they become problems. This section lists the primary maintenance metrics alphabetically with a description and formula as prescribed in AFI 21-103, Equipment Inventory, Status and Utilization Reporting, Attachment 2 (for maintenance status codes); and TO 00-20-2, Maintenance Data Documentation, Appendix L (for flying codes).

A2.3.1. Maintenance Metrics Algorithms. Use *T.O. 00-20-2* as the official source for all maintenance metrics unless otherwise indicated.

A2.3.2. AFMC Specific Algorithms.

A2.3.2.1. Mission Effectiveness Rate (ME)

A2.3.2.1.1. **((TEST MISSIONS SCHEDULED + TNG SORTIES SCHEDULED)** – (NON-EFFECTIVE TEST + NON-EFFECTIVE TNG)) / ((TEST MISSIONS SCHEDULED + TNG SORTIES SCHEDULED) – (TEST MISSIONS NON-EFFECTIVE FOR WEATHER + TNG SORTIES NON-EFFECTIVE FOR WEATHER)) * 100

A2.3.2.2. Test Sortie Generation Rate

A2.3.2.2.1. **(TEST SORTIES FLOWN + GROUND TEST EXECUTED)** / (SCHEDULED TEST SORTIES) * 100

A2.3.2.3. Support Sortie Generation Rate

A2.3.2.3.1. **(SUPPORT SORTIES FLOWN)** / (SCHEDULED SUPPORT SORTIES) * 100

A2.3.2.4. Test Pilot School Sortie Generation Rate

A2.3.2.4.1. (TEST PILOT SCHOOL SORTIES FLOWN) / (SCHEDULED TEST PILOT SCHOOL SORTIES) * 100

A2.3.2.5. Test Pilot School Total Requirement Rate

A2.3.2.5.1. **(SCHEDULED TEST PILOT SCHOOL SORTIES)** / (TOTAL TEST PILOT SCHOOL SORTIES REQUESTED) * 100

A2.3.2.6. Proficiency Sortie Generation Rate

A2.3.2.6.1. (PROFICIENCY SORTIES FLOWN)/(SCHEDULED PROFICIENCY SORTIES) * 100

A2.3.2.7. TSE Rate

A2.3.2.7.1. **(TOTAL GROUND AND SORTIE EVENTS SCHEDULED – TOTAL DEVIATIONS)** / (TOTAL GROUND AND SORTIE EVENTS SCHEDULED) * 100